

	9	3	0	(7, 2)	$x = x+1, y = y+1, p = p+2\Delta y - 2\Delta x$	$ p-13$
12	6					
	10	4	-22	(10, 4)	$x = x+1, p = p+2\Delta y$	
	11	4	-16	(11, 4)	"	
	12	4	-10	(12, 4)	"	
	13	4	-4	(13, 4)	"	
	14	4	2	(14, 4)	$x = x+1, y = y+1, p = p+2\Delta y - 2\Delta x$	
	15	5	-20	(15, 5)	$x = x+1, p = p+2\Delta y$	
	16	5	-14	(16, 5)	"	
	17	5	-8	(17, 5)	"	

$$y = mx + c$$

$$y = m(x_k + 1) + c$$

$$d_1 = y - y_k = m(x_k + 1) + c - y_k$$

$$d_2 = (y_{k+1} - y) = y_{k+1} - m(x_k + 1) - c$$

$$d_1 - d_2 = 2m(x_k + 1) - 2y_k + 2c - 1$$

$$p_k = \Delta x (d_1 - d_2)$$

$$= 2\Delta y \cdot x_k - 2\Delta x \cdot y_k + b$$

$$p_{k+1} = 2\Delta y \cdot x_{k+1} - 2\Delta x \cdot y_{k+1} + b$$

$$= p_k + 2\Delta y - 2\Delta x (y_{k+1} - y_k) \quad \{x_{k+1} = x_k + 1\}$$

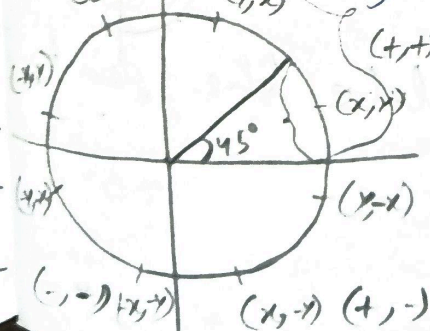
(this value could be 0/1)

Drawing Circle

(-,-)

(-x)

$$(x, y) \quad (x - x_c)^2 + (y - y_c)^2 = r^2$$



We have to draw 1/8th portion of the circle.

$f_{\text{circle}}(x, y) \begin{cases} < 0 \Rightarrow \text{Point is inside the circle.} \\ = 0 \Rightarrow \text{Point is on the surface of the circle.} \\ > 0 \Rightarrow \text{Point is outside the circle.} \end{cases}$

$$p_k = f_{\text{circle}}(x_{k+1}, y_k - \frac{1}{2})$$

$$P_{k+1} = f_{\text{circle}}(x_{k+1} + 1, y_{k+1} - \frac{1}{2})$$

$$P_{k+1} = P_k + 2(x_{k+1} + 1) + (y_{k+1}^2 - y_k^2) - (y_{k+1} - y_k) + 1$$

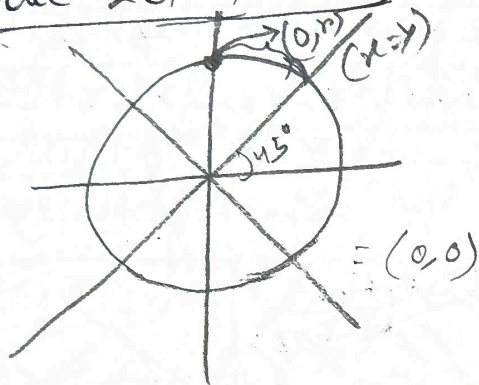
Date - 26/02/2026

$$(1, r) \text{ or } (1, r-1)$$

Formula

$$\left\{ p = \frac{5}{4} - r = 1 - r \right.$$

$$p = 3 - 2r$$



Midpoint Circle Drawing

Mathematical Algo 10-

Algorithm (Computer Graphics C Version Page 111-112)

#include "device.h"

void circleMidpoint (int xCentre, int yCentre, int radius)

```
{
    int x=0;
    int y=radius;
    int p=1-radius;
    void circlePlot points;
```

```
    setPixel (xCentre - x, yCentre - x);
```

};

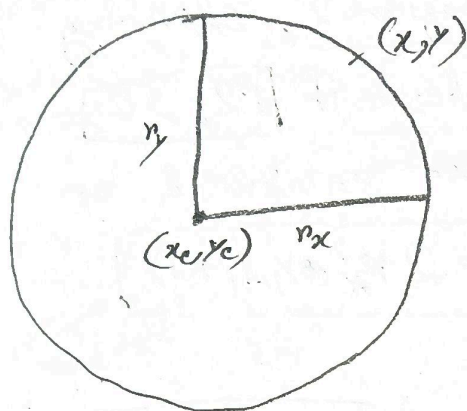
Problem:-
Centre is (0,0) radius is 10 calculate the points.

Date- 11/03/2026

5th Class

P-15

Midpoint Ellipse:-



$$\left(\frac{x-x_c}{r_x}\right)^2 + \left(\frac{y-y_c}{r_y}\right)^2 = 1$$

Algorithm:- for region 1

Step-1:- Input r_x, r_y & ellipse centre (x_c, y_c) & obtain the first point on an ellipse centered on the origin

as $(x_0, y_0) = (0, r_y)$

Step-2:- Calculate the initial value of the decision parametre in region 1 as $P_0 = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2$

Step-3:- At each x_k position in region 1 starting at $k=0$, perform the following step if $P_k < 0$, the next point along the ellipse centered on $(0,0)$ is (x_{k+1}, y_k) & $P_{k+1} = P_k + 2r_y^2 x_{k+1} + r_y^2$, otherwise the next point along the ellipse is (x_{k+1}, y_{k+1}) &

$$P_{k+1} = P_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2$$

Algorithm for Region-2

Step-4:- Calculate the initial value of the decision parametre in region 2 using the last point (x_0, y_0)

calculated in region 1 as ~~P_2~~

$$P_{20} = r_y^2 - (x_0 + \frac{1}{2})^2 + r_x^2 (y_0 - 1)^2 - r_x^2 r_y^2$$

Step-5: At each y_k position in region 2, starting at $[k=0]$ perform the following test if $[P_{2k} > 0]$, the next point along the ellipse centered on $(0,0)$ is $[x_k, y_{k-1}]$ & $P_{2k+1} = P_{2k} + 2r_x^2 x_{k+1} + r_x^2$

otherwise;

the next point along the ellipse is $[x_{k+1}, y_{k-1}]$

$$P_{2k+1} = P_{2k} + 2r_x^2 x_{k+1} - 2r_x^2 y_{k+1} + r_x^2$$

Step-6: Determine symmetric points in the other three co-ordinates.

Step-7: Move each calculated pixel position (x, y) on to the elliptical path centred on (x_c, y_c) and ~~the~~ plot the co-ordinate values $[x = x + x_c]$,

$$[y = y + y_c]$$

Step-8: Repeat the steps for region 1 until

$$2r_y^2 x \geq 2r_x^2 y$$

P-16

Ex

Sup

you

Solut

P16

C

K

0

1

2

3

4

5

6

For

The

H.W

Deriv

① L

② C

③ E

Pr Expl Problem

Suppose $r_x = 8$ & $r_y = 6$ and (x_0, y_0) given as $(0, 0)$ now you have to generate the points on co-ordinate 1.

Solution :- $(x_0, y_0) = (0, 6)$

$$P_{10} = 36 - 64 \times 6 + \frac{1}{4} 64 = -332$$

~~PR~~

For region 1

K	P_{1k}	(x_{k+1}, y_{k+1})	$2r_y^2 x_{k+1}$	$2r_x^2 y_{k+1}$
0	-332	(1, 6)	72	768
1	-224	(2, 6)	144	768
2	-44	(3, 6)		
3		(4, 6)		
4		(5, 5)		
5		(6, 4)		
6		(7, 3)		

For region 2

The points are gonna be $(8, 2)$, $(8, 1)$, $(8, 0)$.

H.W

Derive equations for decision parameters update.

- ① Line Drawing
- ② Circle Drawing
- ③ Ellipse Drawing

Date-19/03/26

Digital Logic

P-18

For Practical of Architecture

Basic Gates, Universal Gates, ~~Half Adder~~ ~~Subtractor~~
Subtractor, Multiplexor, de-multiplexor, Decoder, Encoder

Combinational Circuits.

vs
Sequential Circuits

SR JK D T Counter Register

* All Circuit Diagrams & Truth Table.

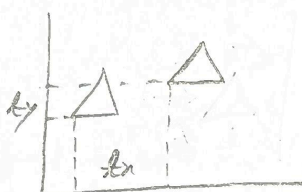
2D Geometric Transformation

① Translation

$$x' = x + tx$$

$$y' = y + ty$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} tx \\ ty \end{bmatrix}$$



Parameters

tx

ty

② Rotation

$$x' = r \cos(\phi + \theta)$$

$$= x \cos \theta - y \sin \theta$$

$$y' = r \sin(\phi + \theta)$$

$$= x \sin \theta + y \cos \theta$$

$$(x', y') = (r \cos(\phi + \theta), r \sin(\phi + \theta))$$

$$(x, y) = (r \cos \theta, r \sin \theta)$$

* Polar Co-ordinates:- (x, y)

$$x = r \cos \theta$$

$$y = r \sin \theta$$

~~***~~

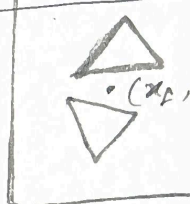
$x' =$

$y' =$

Matrix Form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \dots \\ \dots \end{bmatrix}$$

Solve this pr



$$x' = x_p + (x - x_p) \cos \theta - (y - y_p) \sin \theta$$

$$y' = y_p + (x - x_p) \sin \theta + (y - y_p) \cos \theta$$

③ Scaling

$$x' = S_x \cdot x$$

$$y' = S_y \cdot y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

~~***~~

$$x' = r \cos(\phi + \theta)$$

$$= r(\cos \phi \cos \theta - \sin \phi \sin \theta)$$

$$= r \cos \phi \cos \theta - r \sin \phi \sin \theta$$

$$= x \cos \theta - y \sin \theta$$

$$y' = r \sin(\phi + \theta)$$

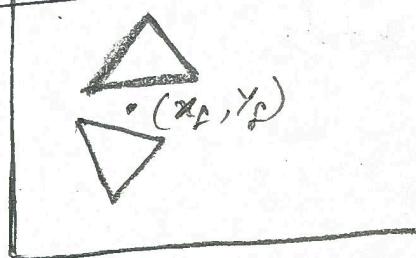
$$= r(\sin \phi \cos \theta + \cos \phi \sin \theta)$$

$$= x \sin \theta + y \cos \theta$$

Matrix form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Solve this problem



$$x' = x_f + (x - x_f) \cos \theta - (y - y_f) \sin \theta$$

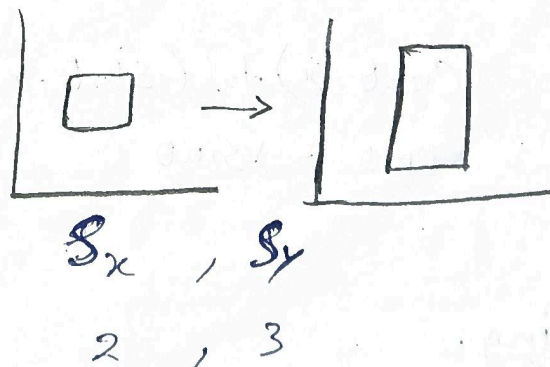
$$y' = y_f + (x - x_f) \sin \theta + (y - y_f) \cos \theta$$

③ Scaling

$$x' = S_x \cdot x$$

$$y' = S_y \cdot y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} S_x \\ S_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$



Homogenous Coordinate system -

K20

$$(x_h, y_h, h)$$

$$\left. \begin{aligned} x &= \frac{x_h}{h} \\ y &= \frac{y_h}{h} \end{aligned} \right\} \rightarrow (x, y, 1)$$

~~Scaling~~ - Translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$x' = 1 \cdot x + 0 \cdot y + t_x \cdot 1$$

$$y' = 1 \cdot y + t_y \cdot 1$$
$$= x + t_x$$

$$x' = x + t_x$$

Rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$x' = (\cos \theta \cdot x) + (-\sin \theta \cdot y) + (0 \cdot 1)$$
$$= x \cos \theta - y \sin \theta$$

$$y' = (\sin \theta \cdot x) + (\cos \theta \cdot y) + (0 \cdot 1)$$
$$= x \sin \theta + y \cos \theta$$

Scaling -

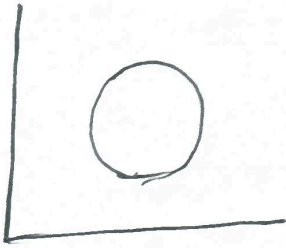
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$x' = s_x \cdot x$$

$$y' = s_y \cdot y$$

Composite Transformation

P-21



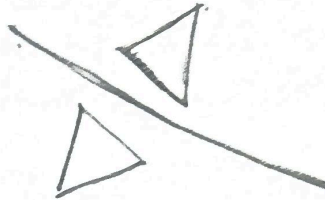
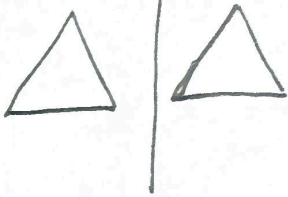
$$t_x = 4$$

$$t_y = 5$$

$$s_x = 2$$

$$s_y = 3$$

Mirror



Transformation Like Shear

